

H2O Online BA End-to-End Guide

Complete Business Analyst reference for the H2O Online water delivery platform — from business need and requirement gathering through analysis, BRD/FRD documentation, process mapping, validation, development & testing support, UAT, deployment, and post go-live. Includes data validation methodology, RTM, interview preparation, and stakeholder communication guides.

12-phase BA lifecycle:

- Business Need → Gathering → Analysis → Process Mapping → Solution Design
- Documentation (BRD, FRD, RTM) → Validation → Dev & Test Support
- UAT → Deployment → Post Go-Live + 4-layer data validation

12

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Version: 1.0

Audience: Business Analysts, PO, QA Leads

Project: H2O Online Platform

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H2O Online — Business Analyst End-to-End Guide

Document type: BA Playbook · Interview Preparation · Process & Methodology Guide

Version: 1.0

Audience: Business Analysts, Product Owners, QA Leads, Delivery Managers

Project: H2O Online — Multi-sided Water Delivery Platform

How to Use This Guide

This document is the **single reference for a Business Analyst** working on H2O Online. Use it to:

- 1. **Explain the project** to stakeholders in interviews or kickoff meetings
- 2. **Follow the end-to-end BA lifecycle** from business need through post go-live
- 3. **Know which artifacts to create** (BRD, FRD, process maps, RTM, UAT pack)
- 4. **Run validation** — functional, data, statistical, and analytics layers
- 5. **Support testing and UAT** with traceable acceptance criteria

Related project documents:

Document	BA use
docs/requirement-capture.md	Master FRD-style requirement catalogue
docs/BUSINESS-USE-CASE-PUNE.md	City launch & operations context
docs/SYSTEM-ARCHITECTURE.md	Technical flows for solution alignment
docs/customer-dashboard-devices-leaderboard-brd.md	Sample feature-level BRD

Part A — Explaining the Project (Interview & Stakeholder Prep)

A.1 The 30-Second Pitch

> **H2O Online** is a multi-sided water delivery platform. Customers order water jars and subscriptions on a mobile app. Verified **suppliers** fulfil orders and assign **riders**. An **admin portal** runs the business — partner verification, service zones by PIN code, stock, wallets, and financials. The platform earns through **supplier commission** (~20%) and manages **rider payouts** (~10% of order value). Beyond delivery, it includes **wallet payments**, **Groq AI hydration insights**, and **Google Maps** for ETA and serviceability.

A.2 The 2-Minute Stakeholder Explanation

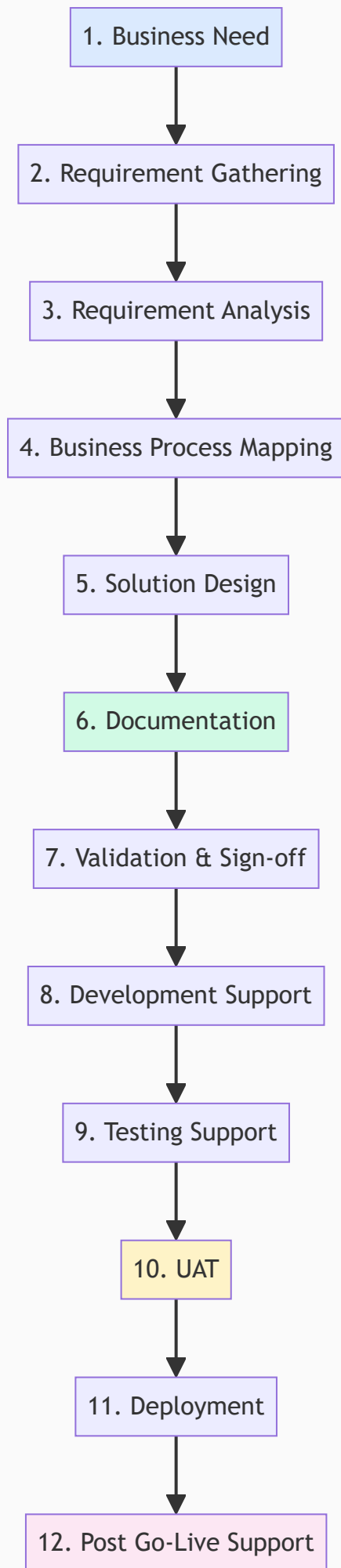
Question stakeholders ask	BA answer (H2O Online)
What problem does it solve?	Fragmented phone/WhatsApp water ordering — no tracking, no subscriptions, no verified supply chain
Who uses it?	Customers, water suppliers, delivery riders, society/corporate buyers (partial), admin ops team
How does money flow?	Customer pays (wallet/Razorpay/COD) → on delivery, platform splits to supplier wallet + rider wallet minus commission
What makes it different?	Verification gate, PIN-based serviceability, stock at order time, subscription billing, AI engagement
What's live vs planned?	Customer/supplier/rider/admin flows live; corporate demo UI; coupons/OTP/live map tracking partial
What would a BA own?	Requirements from stakeholders, process maps, BRD/FRD, acceptance criteria, UAT, data validation rules

A.3 Five Things a BA Must Internalise

- 1. **Multi-sided marketplace** — requirements must be written **per actor** (customer, supplier, rider, admin)
- 2. **Order is the core transaction** — stock deduct, serviceability, payout on deliver
- 3. **Admin is the control tower** — verification, zones, financials, support
- 4. **Not everything in the UI is implemented** — BA must maintain **As-Is vs To-Be** gap list
- 5. **Data integrity matters** — wallet balances, stock counts, commission math must reconcile

DIAGRAM 1





B.1 BA Lifecycle — Phase Summary

Phase	BA primary outputs	H2O Online focus
1. Business Need	Problem statement, business case, objectives	BO-01 to BO-07 from requirement capture
2. Requirement Gathering	Interview notes, workshop outputs, stakeholder register	5 personas + admin roles
3. Requirement Analysis	Prioritised backlog, MoSCoW, gap analysis	Live vs partial features
4. Business Process Mapping	As-Is / To-Be BPMN, swimlanes	Order, subscription, onboarding flows
5. Solution Design	BRD sections, business rules, wireframe briefs	Mobile + admin + API alignment
6. Documentation	BRD, FRD, user stories, RTM	FR-xxx IDs, BR-xxx rules
7. Validation & Sign-off	Review pack, signed BRD/FRD	PO + Sponsor sign-off
8. Development Support	Clarifications, change requests	API ↔ screen traceability
9. Testing Support	Test scenarios, expected results	Per FR ID
10. UAT	UAT scripts, defect log, sign-off	Role-based UAT packs
11. Deployment	Release notes, training, go-live checklist	AWS prod, Pune launch
12. Post Go-Live	Hypercare log, enhancement backlog	KPI monitoring, data reconciliation

Part C — Phase-by-Phase BA Playbook

Phase 1 — Business Need

Objective: Confirm *why* the project exists and what success looks like.

BA activities

Activity	Deliverable	H2O Online example
Identify business problem	Problem statement	Offline water ordering lacks tracking, subscriptions, trust
Define business objectives	Objective table (BO-xx)	BO-01 order journey, BO-02 subscriptions, BO-03 partner scale
Identify stakeholders	Stakeholder register	Product owner, ops head, suppliers, riders, customers, dev lead
Assess current state	As-Is summary	Phone orders, cash, no central admin
Define success metrics	KPI list	Orders/week, fulfilment %, subscription conversion
Scope boundaries	In / out of scope	Corporate demo = out; customer order = in

Stakeholder register template

Stakeholder	Role	Interest	Influence	BA engagement
Product Owner	Sponsor	ROI, roadmap	High	Weekly backlog review
City Ops Manager	Admin user	Pune launch	High	Process workshops
Supplier rep	Supply side	Orders, payouts	Medium	Journey walkthrough
Delivery lead	Fulfilment	Rider assignment	Medium	SLA definition
Customer support	Operations	Tickets	Medium	Support workflow FRs
Tech Lead	Delivery	Feasibility	High	FR review, API mapping
QA Lead	Quality	Test coverage	Medium	RTM, UAT plan

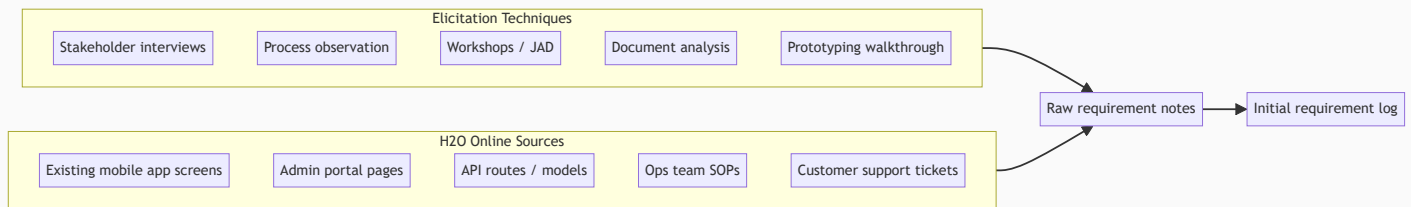
Business objectives (reference — from project)

ID	Objective	BA traceability
BO-01	On-demand + scheduled ordering	FR-CUS-10 to FR-CUS-29, order workflow
BO-02	Subscription recurring revenue	FR-CUS-37 to FR-CUS-44, Workflow B
BO-03	Verified partner onboarding	FR-SUP-12, FR-DEL-13, FR-ADM-06 to 11
BO-04	Transparent partner economics	FR-SUP-09/10, FR-DEL-09, BR-05
BO-05	Central admin operations	FR-ADM-01 to FR-ADM-27
BO-06	Engagement beyond ordering	FR-CUS-45 to 49, AI features
BO-07	Trust via verification + support	FR-CUS-50 to 52, BR-01

Phase 2 — Requirement Gathering

Objective: Elicit needs from all actors without solution bias.

DIAGRAM 2



Requirement gathering plan by actor

Actor	Key questions	Sessions	Output
Customer	How do you choose supplier? Address pain? Payment preference?	2 interviews + app walkthrough	Customer journey FRs
Supplier	Accept/reject rules? Stock update frequency? Rider model?	2 interviews	Supplier FRs + BRs
Rider	Online hours? Pickup proof? Earnings visibility?	1 interview	Delivery FRs
Admin	Verification checklist? Financial access? Zone setup?	2 workshops	Admin FRs + RBAC matrix
Product Owner	Priority? Launch city? Revenue model?	Ongoing	Prioritised backlog

Requirement gathering checklist

- ☐ Stakeholder register approved
- ☐ Interview guide per persona completed
- ☐ As-Is process documented (phone order flow)
- ☐ Demo of mobile app (customer, supplier, rider paths)
- ☐ Demo of admin portal (all nav groups)
- ☐ Review `requirement-capture.md` for baseline
- ☐ Gap list: UI-only vs backend-implemented
- ☐ Non-functional needs captured (SLA, security, privacy)
- ☐ Assumptions and constraints log started

Sample interview questions — Customer

- Walk me through your last water order outside the app.
- What information do you need while waiting for delivery?
- Would you prepay via wallet? What would make you trust it?
- How important are subscriptions vs one-time orders?
- What would make you abandon checkout? (address, stock, payment)

Sample interview questions — Admin Ops

- How do you decide if a supplier is approved?
- How do you define which PIN codes are serviceable?
- What do you do when a supplier rejects an order?
- How do you reconcile rider and supplier payouts weekly?
- Which reports do you need on day 1 of Pune launch?

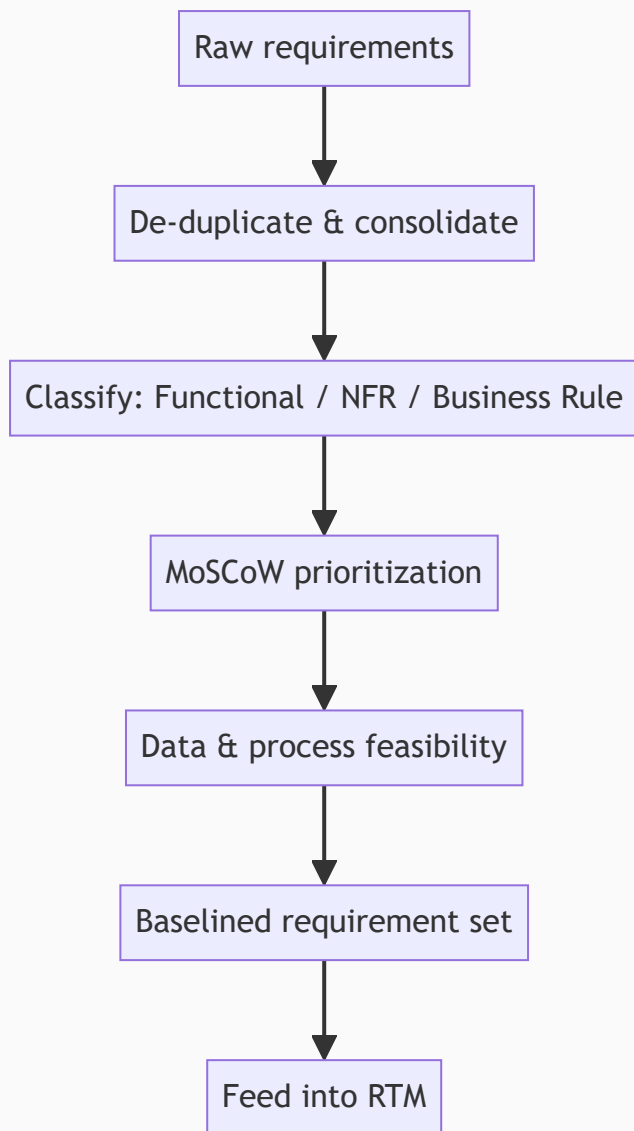
Phase 3 — Requirement Analysis

Objective: Structure, prioritise, and validate requirements for consistency and feasibility.

Analysis techniques on this project

Technique	Application
MoSCoW	Must = order + pay + deliver; Should = AI insights; Could = coupons
Gap analysis	Compare UI (coupon field) vs API (not implemented)
Dependency mapping	Serviceability before checkout; verify before supplier dashboard
Conflict resolution	Sub-admin wants financials vs BR-06 RBAC
Feasibility review	Tech lead confirms Razorpay verify flow vs direct order
Impact analysis	Change to commission % affects supplier financials + admin reports

DIAGRAM 3



Requirement types and ID scheme

Type	Prefix	Example	Document
Business objective	BO-xx	BO-02 subscriptions	BRD \$2
Business rule	BR-xx	BR-01 verify before transact	BRD \$Business Rules
Functional requirement	FR-xxx-yy	FR-CUS-18 select address	FRD
Non-functional	NFR-xx	NFR-05 real-time tracking	BRD \$NFR
User story	US-xxx	As a customer I want...	Backlog / sprint
Acceptance criterion	AC-xxx	Given... When... Then...	FRD / UAT
Backlog item	BL-xx	BL-04 coupon engine	Future scope

Gap analysis template (H2O Online — examples)

Feature (UI)	Backend/API	Gap	BA recommendation
Coupon at checkout	Not implemented	High	BL-04; remove from Must FR or mark Could
OTP login	Mock flow	High	BL-05; document as non-prod
Live map on track screen	Partial / placeholder	Medium	BL-06; UAT uses status timeline only
Loyalty points display	Display only	Low	Document as cosmetic until BL-03
Razorpay pay	Implemented verify flow	None	FR-CUS-23 updated to include Razorpay path
Stock check	Implemented on order	None	Include in UAT stock scenarios

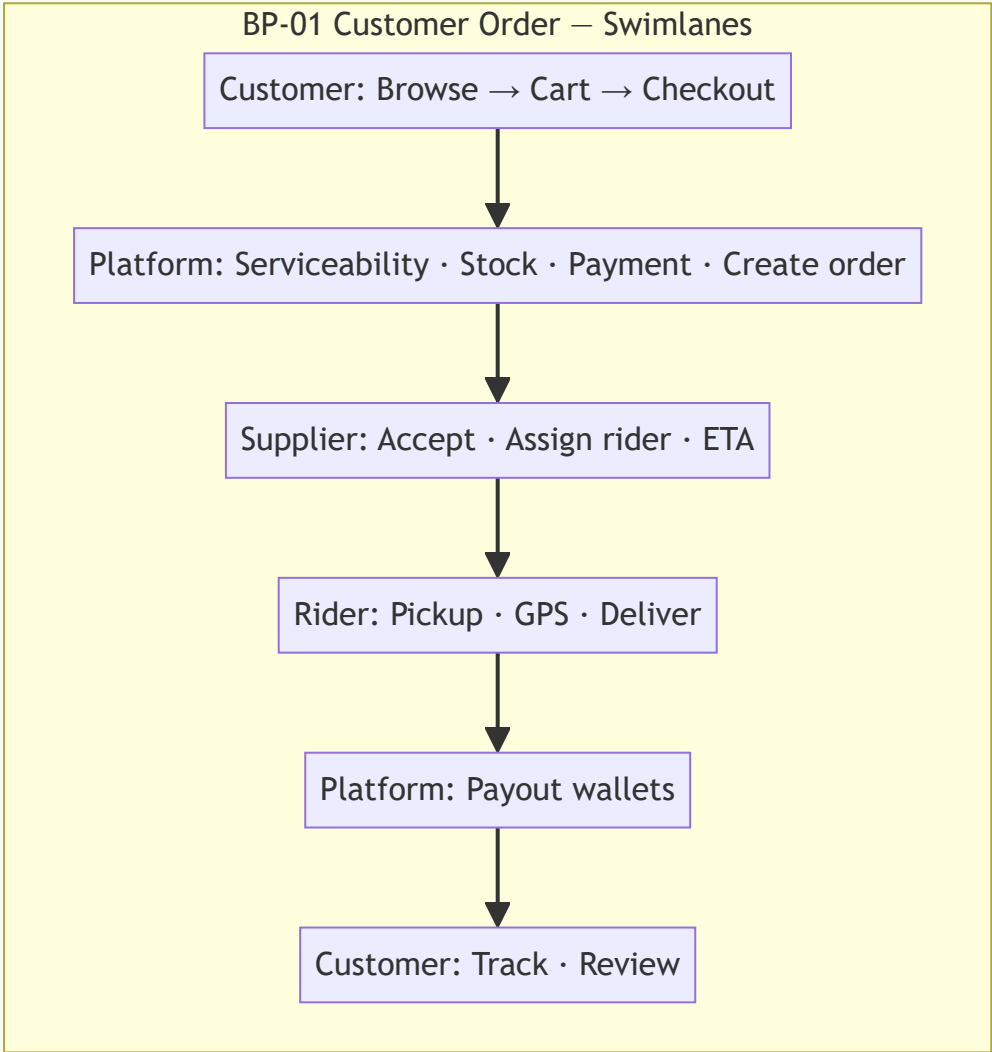
Phase 4 — Business Process Mapping

Objective: Visualise how work flows across people, systems, and data.

Processes the BA must map (minimum set)

Process ID	Name	Actors	Priority
BP-01	Customer one-time order	Customer, Platform, Supplier, Rider	Critical
BP-02	Subscription lifecycle	Customer, Admin, Rider	Critical
BP-03	Supplier onboarding	Supplier, Admin	Critical
BP-04	Rider onboarding & fulfilment	Rider, Supplier, Admin	Critical
BP-05	Store approval	Supplier, Admin	High
BP-06	Serviceability & address	Customer, Admin, Google Maps	High
BP-07	Payment & wallet settlement	Customer, Platform, Supplier, Rider	Critical
BP-08	Customer support ticket	Customer, Admin	Medium
BP-09	Admin daily operations	Admin, Sub-admin	High

DIAGRAM 4



As-Is vs To-Be template

Step	As-Is (offline Pune vendor)	To-Be (H2O Online)	System touchpoint
Order	Phone call / WhatsApp	App checkout	POST /api/orders
Address	Verbal	Saved address + PIN + map	/api/addresses
Payment	Cash on delivery	Wallet / Razorpay / COD	payments + wallet
Assign delivery	Vendor's own staff	Supplier assigns rider in app	supplier orders API
Track	Call vendor	Order tracking API	/api/orders/id/tracking

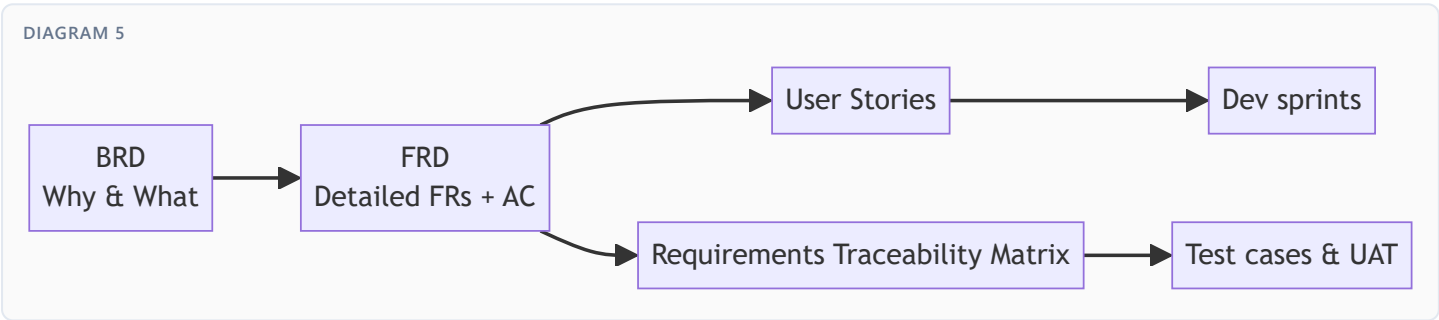
Step	As-Is (offline Pune vendor)	To-Be (H2O Online)	System touchpoint
Complaint	Phone	Support ticket	/api/customer-support

Phase 5 — Solution Design (BA Contribution)

Objective: Translate business needs into solution boundaries the dev team can build.

Note: Solution *architecture* is owned by Tech Lead; BA contributes **business solution design**.

BA deliverable	Content
Solution scope diagram	Mobile + Admin + API + MongoDB + 3rd parties
Business rules catalogue	BR-01 to BR-10
Entity glossary	User, Order, Supplier, Store, ServiceableArea, Wallet
Integration list	Groq, Google Maps, Razorpay, MongoDB Atlas
Role-permission matrix	Admin RBAC
Wireframe / screen inventory	Reference docs/screens-catalog.json
API-to-requirement map	Which FR maps to which endpoint



Phase 6 — Documentation

Objective: Produce controlled, versioned artifacts stakeholders can sign off.

Document hierarchy

Document	Purpose	Owner	Audience
BRD (Business Requirements Document)	Business need, objectives, scope, processes, rules, NFRs	BA	Sponsor, PO, Ops
FRD (Functional Requirements Document)	Detailed FR-IDs, acceptance criteria, screen behaviour	BA	Dev, QA, PO
RTM (Requirements Traceability Matrix)	FR → Design → Test → UAT link	BA	QA, PM
Process maps	BPMN / swimlanes	BA	All stakeholders
Data dictionary	Field definitions for orders, wallet, stock	BA + Tech	Dev, analytics
UAT pack	Scripts per role	BA + QA	Business users
Release notes	What's in this release	BA + Dev	Ops, support

BRD — Standard sections (H2O Online template)

- 1. **Executive summary** — 1 page business narrative
- 2. **Business objectives** — BO-01 to BO-07 with KPIs
- 3. **Stakeholders & personas** — customer, supplier, rider, admin
- 4. **Scope** — in scope / out of scope / assumptions
- 5. **Business process overview** — BP-01 to BP-09 summaries
- 6. **Business rules** — BR-01 to BR-10
- 7. **Business capabilities** — wallet, commission, AI, support
- 8. **Non-functional requirements** — NFR-01 to NFR-10
- 9. **Financial / commercial rules** — commission 20%, rider 10%
- 10. **Risks & dependencies** — Razorpay, Maps API, verification SLA
- 11. **Success criteria & sign-off** — measurable outcomes

Existing BRD example: docs/customer-dashboard-devices-leaderboard-brd.md

FRD — Standard sections (H2O Online template)

1. **Introduction & references** — link to BRD
2. **Functional requirements by module**
- FR-AUTH-xx Authentication

- FR-CUS-xx Customer

- FR-SUP-xx Supplier

- FR-DEL-xx Delivery partner

- FR-ADM-xx Admin
1. **Acceptance criteria** per FR (Given/When/Then)
2. **UI screen mapping** — screen name → FR IDs
3. **API mapping** — endpoint → FR IDs
4. **Error & edge cases** — stock-out, unserviceable PIN, reject order
5. **Open issues & TBD log**

Master FR catalogue: docs/requirement-capture.md (use as FRD baseline)

User story format (for backlog)

As a [supplier in Pune]
I want to [accept an incoming order and assign a rider]
So that [the customer receives delivery within SLA]

Acceptance criteria:

- AC-1: Given pending order, when supplier taps Accept, then status = accepted

- AC-2: Given accepted order, when rider selected, then deliveryPartnerId saved

- AC-3: Given no riders online, then UI shows warning (partnerAvailability)

Linked: FR-SUP-04, FR-SUP-06, BP-01

Requirements Traceability Matrix (RTM) — sample rows

FR ID	BRD objective	Process	API / Component	Test case	UAT script
FR-CUS-18	BO-01	BP-01	CheckoutScreen, /api/addresses	TC-ORD-012	UAT-C-05
FR-CUS-18	BO-01	BP-06	/api/serviceability/check	TC-ORD-013	UAT-C-06
FR-SUP-04	BO-01	BP-01	PATCH .../orders/:id/accept	TC-SUP-003	UAT-S-02
FR-DEL-03	BO-01	BP-04	PATCH .../picked-up	TC-RID-002	UAT-R-02
FR-ADM-06	BO-03	BP-03	Admin Suppliers verify	TC-ADM-004	UAT-A-03
FR-ADM-20	BO-05	BP-07	Wallet management	TC-ADM-010	UAT-A-08

Phase 7 — Validation & Sign-off

Objective: Confirm requirements are complete, correct, and agreed before build/test at scale.

Review type	Participants	BA role	Exit criteria
Business review	PO, Ops, Sponsor	Present BRD	Objectives & scope signed
Functional review	PO, Dev, QA	Present FRD + RTM	No open Must gaps
Technical review	Tech Lead, Dev	Clarify API mapping	Feasibility confirmed
Compliance review	Legal / PO	Privacy, GST, FSSAI notes	NFR sign-off

Sign-off checklist

- [] All Must-have FRs have acceptance criteria
- [] Business rules validated with Ops (commission, verification)
- [] Out-of-scope items explicitly listed (coupons, OTP)
- [] RTM covers all Must FRs
- [] Assumptions documented and accepted
- [] Sign-off sheet completed (see requirement-capture.md §18)

Phase 8 — Development Support

Objective: Answer clarifications and manage controlled change during build.

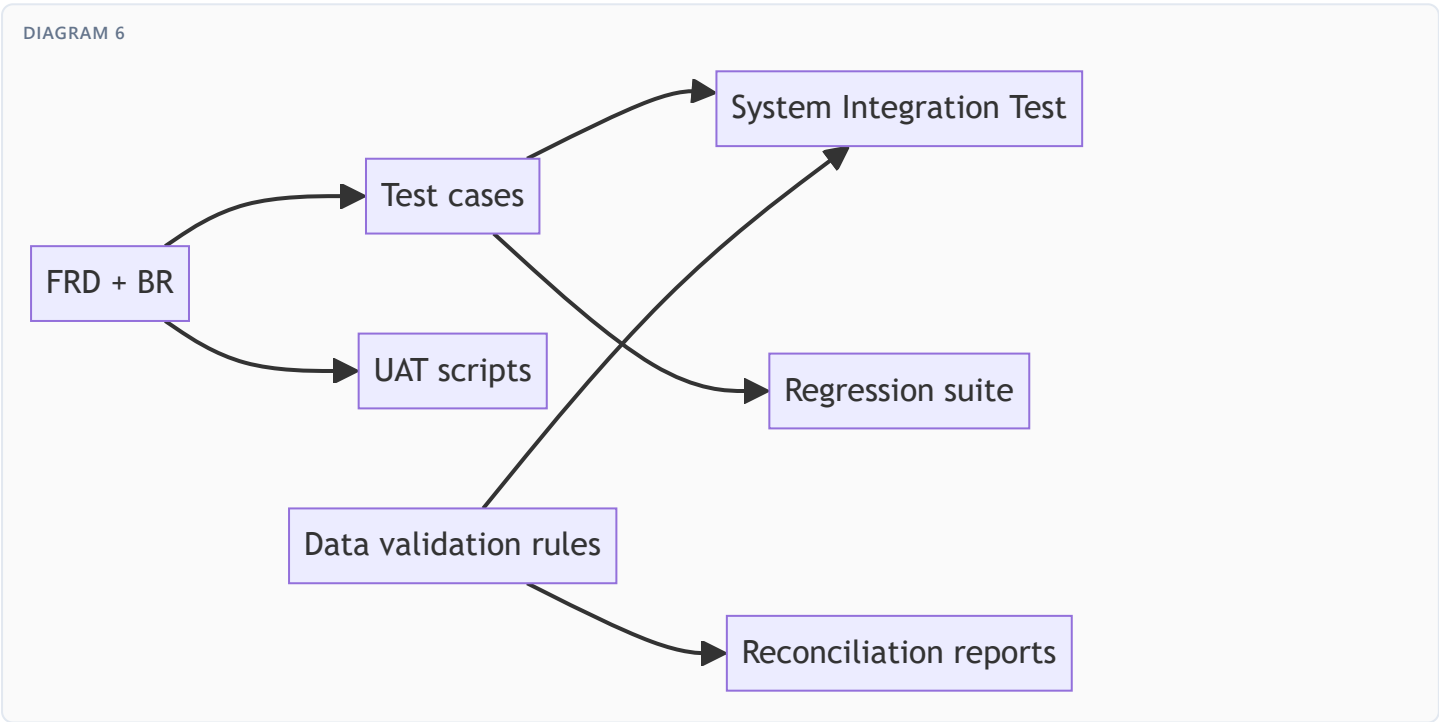
Activity	BA action	Tool / artifact
Clarification requests	Respond within SLA; update FR if needed	CR log
Sprint refinement	Explain user story context	Backlog
API contract review	Map request/response to FR	Postman / OpenAPI notes
Change request	Impact: FR, RTM, test, UAT	Change impact matrix
Demo attendance	Verify against acceptance criteria	Sprint demo checklist

Change impact matrix template

Change	Affected FRs	Processes	Test impact	UAT impact	Approval
Add PIN radius 15km	FR-ADM serviceable areas	BP-06	TC-SVC-*	UAT-A-12	PO

Phase 9 — Testing Support

Objective: Ensure QA can trace tests to requirements and business rules.



Test scenario categories (H2O Online)

Category	Examples	Linked rules
Happy path	End-to-end order delivered	BP-01
Negative	Stock-out, invalid PIN, wrong password	BR + validations
RBAC	Sub-admin blocked from financials	BR-06, FR-ADM-23
Integration	Razorpay verify → order create	Payment flow
Boundary	stockQty = 0, radius edge PIN	Serviceability
Concurrency	Two orders last jar	Stock deduct
Financial	Payout math on deliver	BR-05, 10% rider

BA test support deliverables

- 1. **Test scenario document** — business scenarios in plain language
- 2. **Expected result per FR** — from acceptance criteria
- 3. **Test data spec** — users, suppliers, PINs, products, stock levels
- 4. **Defect triage rules** — Must FR failure = blocker

Phase 10 — UAT (User Acceptance Testing)

Objective: Business users confirm the system meets agreed requirements before go-live.

UAT methodology for H2O Online

Step	Activity	Owner
1	Define UAT scope (Must FRs only for pilot)	BA + PO
2	Create role-based UAT packs	BA
3	Prepare UAT environment + test data	QA + Ops
4	Train UAT participants (30 min per role)	BA
5	Execute UAT scripts; log defects	Business users
6	BA triages defects: bug vs change vs training	BA
7	Re-test fixes	Business users
8	UAT sign-off memo	PO + Ops

UAT packs by role

Pack	Tester	Critical scripts
UAT-C Customer	Ops team member	Register, order, pay, track, support ticket
UAT-S Supplier	Pilot vendor	Onboarding, accept order, assign rider, stock update
UAT-R Rider	Pilot rider	Go online, pickup, deliver, view earnings
UAT-A Admin	City manager	Verify supplier, serviceable PIN, orders dashboard, wallet adjust

Sample UAT script — UAT-C-05 (Address & serviceability)

Step	Action	Expected result	FR ref	Pass/Fail
1	Login as customer	Dashboard loads	FR-AUTH-06	
2	Add address Kothrud PIN 411038 via map	Address saved	FR-CUS-30/31	
3	Add product to cart → checkout	Address pre-selected	FR-CUS-18	
4	Proceed with serviceable PIN	Checkout allowed	FR-CUS-18, BP-06	
5	Change PIN to invalid 000000	Serviceability error shown	BP-06	
6	Complete order with wallet	Order confirmed	FR-CUS-23/24	

UAT entry / exit criteria

Entry: FRD signed off; SIT complete; critical defects = 0; UAT data loaded

Exit: 100% Must UAT scripts pass; PO sign-off; known issues documented with workaround

Phase 11 — Deployment

Objective: BA ensures business readiness for production go-live.

BA deliverable	Content
Go-live checklist	Admin users created, suppliers verified, PINs configured
Release notes	Features enabled; known limitations (coupons off)
Training materials	Admin 2hr workshop; supplier quick guide
Support playbook	Escalation paths, SLAs
Rollback criteria	Payment failures > X%; API down > Y min

Reference: docs/DEPLOYMENT-GUIDE.md , docs/BUSINESS-USE-CASE-PUNE.md

Phase 12 — Post Go-Live Support

Objective: Hypercare, measure KPIs, feed enhancements back to backlog.

Activity	Frequency	BA output
Hypercare standup	Daily (week 1)	Issue log
KPI review	Weekly	Orders, fulfilment %, stock-out rate
Data reconciliation	Weekly	Wallet vs order report
Retrospective	End week 2	Enhancement backlog

Activity	Frequency	BA output
FR gap review	Monthly	Update gap analysis

Part D — Data Validation & Analytics Methodology

BA owns **business data rules**; validation spans multiple layers aligned to H2O Online architecture.

DIAGRAM 7

Layer 1 – Source / Input

Mobile app forms

Admin portal entries

Razorpay webhooks

Google Maps responses



Layer 2 – API / Staging Validation

Request schema validation

Business rule enforcement
stock · PIN · RBAC

Integration smoke tests



Layer 3 – Application DB (MongoDB Atlas)

Entity integrity
User · Order · Product · Wallet

Referential consistency
supplierId · storeId · userId

State machine correctness
order status transitions



Layer 4 — Reconciliation / Analytics

Financial reconciliation

Operational KPIs

Statistical quality checks

D.1 Layer 1 — Source data validation

Data element	Business rule	Validation check
PIN code	6 digits, serviceable	Regex + serviceability API
Email	Unique per user	Register/login
stockQty	Non-negative integer	Product create/update
Order total	subtotal + tax = total	Billing validation
Commission %	0–100	Admin supplier settings
lat/lng	Required for store	Store creation

D.2 Layer 2 — API / staging validation

Staging environment: Mirror of prod schema on separate MongoDB database; same API code (`NODE_ENV=staging`).

Test type	Script / method	BA sign-off focus
Smoke test	<code>backend/scripts/integration-smoke-test.js</code>	Health, auth, serviceability
Negative API tests	Invalid PIN, insufficient stock	Error messages match FR
Payment staging	Razorpay test keys	Verify → order created once
RBAC API tests	Sub-admin → financials = 403	BR-06

D.3 Layer 3 — Application database validation

Entity	Key fields to validate	SQL-like check (Mongo aggregation concept)
Order	status transitions legal	No delivered → in_progress
Order	supplierResponses match items	One response per supplierId in items
Product	stockQty ≥ 0	Count where stockQty < 0 = 0
Product	inStock ↔ stockQty	inStock false when qty = 0
Wallet	balance = sum(transactions)	Reconciliation query
Wallet	payout once per order	No duplicate supplier_payout ref
ServiceableArea	PIN + supplierId unique	No duplicate active zones
User	role matches linked entity	supplier role → Supplier doc exists

Order state validation rules

From	To	Allowed?
in_progress	delivered	Yes (via rider deliver)
in_progress	cancelled	Yes (customer/supplier)
delivered	in_progress	No — data defect
pending supplier	accepted	Yes
accepted	picked_up	Yes
picked_up	delivered	Yes

D.4 Layer 4 — Reconciliation & analytics validation

Financial reconciliation (weekly BA report)

Platform revenue (week) = Sum(order.total × commission%) for delivered orders + platform retention after rider 10% Supplier wallet credits = Sum(transactions where ref starts with supplier_payout_) Rider wallet credits = Sum(transactions where ref starts with delivery_) Check: supplier_payout + rider_payout + platform_cut ≈ order totals (delivered)
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Metric	Formula	Alert threshold
Stock-out rate	Orders failed stock / total attempts	> 3%
Fulfilment rate	Delivered / (Delivered + Cancelled)	< 95%
Avg delivery time	deliveredAt – createdAt	> 45 min Pune pilot
Unserviceable rate	Failed serviceability checks / check calls	Spike > 10% day-over-day
Wallet drift	wallet.balance – computed sum(transactions)	≠ 0
Duplicate payout	Count payout refs per orderId	> 1

Statistical validation checks

Check	Method	Purpose
Order volume anomaly	7-day moving avg ± 2σ	Detect outage or marketing spike
PIN distribution	Orders by pinCode histogram	Validate zone rollout
Cancellation correlation	Cancel rate by supplier, hour	Ops improvement
Stock depletion rate	Δ stockQty vs orders by product	Forecast restock
Subscription churn	Cancelled subs / active subs	BO-02 health
A/B payment mix	wallet vs razorpay vs cod %	Payment strategy

Analytics perspectives BA should define

Perspective	Questions	Data source
Customer	Repeat rate? Avg order value?	Orders by userId
Supplier	Accept time? Reject rate?	supplierResponses
Rider	Deliveries/day? Online hours?	DeliveryPartner + orders
Geographic	Hot PINs? Unserviceable gaps?	Orders + ServiceableArea
Financial	GMV, take rate, payout lag	Orders + Wallets + Admin financials
Quality	Review scores, ticket volume	Reviews + Support tickets

D.5 Data validation sign-off template

Layer	Validator	Date	Status	Notes
L1 Source rules	BA			
L2 API staging	QA			
L3 DB integrity	QA + Dev			

Layer	Validator	Date	Status	Notes
L4 Reconciliation	BA + Finance			
Analytics KPIs	BA + Ops			

Part E — BA Interview Preparation

E.1 "Walk me through this project as a BA"

Structured answer (STAR-style overview):

1. **Situation:** Fragmented water delivery market; need digital platform for Pune-scale ops
2. **Task:** Define requirements across 4 actors + admin; ensure traceability to test/UAT
3. **Action:** Elicited from stakeholders; mapped BP-01 order flow; wrote FR-CUS/SUP/DEL/ADM catalogue; RTM to UAT; data validation on wallet/stock
4. **Result:** Live platform with order → verify → deliver → payout; admin controls zones and partners

E.2 Common interview questions & answers

Question	Strong BA answer (H2O Online)
Who were your stakeholders?	PO, Pune ops (admin), suppliers, riders, customers, tech lead, QA
Hardest requirement conflict?	Sub-admin wanted financials vs BR-06 — resolved via RBAC matrix sign-off
How did you prioritize?	MoSCoW: order path Must; coupons Could (not built)
How do you handle scope creep?	Change impact matrix; PO approves; RTM updated
What artifacts did you produce?	BRD objectives, FRD with FR-IDs, process maps, RTM, UAT packs
How did you validate data?	4 layers: input rules, API staging, MongoDB integrity, wallet reconciliation
Example business rule?	BR-01: supplier cannot transact until admin verifies documents
How does UAT differ from SIT?	SIT = QA technical; UAT = business users confirm FR acceptance criteria
How do you trace requirements?	RTM: FR → API → test case → UAT script
What metrics matter post launch?	Fulfilment rate, stock-out %, subscription conversion, wallet drift

E.3 Whiteboard process — be ready to draw

1. Customer order flow (5 swimlanes)
2. Supplier onboarding → admin verify
3. Money flow: pay → deliver → commission split
4. BA lifecycle from business need to go-live

Part F — Master Checklists

F.1 New BA onboarding (Week 1)

- ☐ Read this guide end-to-end
- ☐ Read `requirement-capture.md`
- ☐ Walk through mobile app: customer, supplier, rider
- ☐ Walk through admin portal all modules
- ☐ Read `SYSTEM-ARCHITECTURE.md` §5–7 (auth, address, order)
- ☐ Run `integration-smoke-test.js` against staging
- ☐ Build stakeholder register for your release
- ☐ Draft RTM skeleton

F.2 Pre-UAT checklist

- ☐ All Must FRs have UAT scripts
- ☐ Test users: 2 customers, 2 suppliers, 3 riders, 3 admin roles
- ☐ Test PINs configured in serviceable areas
- ☐ Products with known stockQty
- ☐ Razorpay test mode documented
- ☐ Known gaps communicated (coupons, OTP)

F.3 Go-live checklist (BA)

- [] BRD/FRD signed
- [] UAT sign-off memo archived
- [] L3 + L4 data validation passed
- [] Training completed for admin ops
- [] Support playbook handed to support team
- [] KPI dashboard defined for week 1 hypercare

Part G — Glossary (BA Quick Reference)

Term	BA definition
Serviceability	Business rule: customer PIN within supplier zone before order allowed
supplierResponses	Per-supplier leg of order: pending → accepted/rejected → delivery stages
Commission	Platform fee % on supplier gross (default 20%)
Delivery share	Rider portion of order total (10%)
RTM	Requirements Traceability Matrix
BRD	Why and what at business level
FRD	Detailed functional behaviour with IDs
Hypercare	Intensive support period after go-live
Gap analysis	UI capability vs backend implementation diff

Document Sign-Off

Role	Name	Date	Signature
Head of Business Analysis			
Product Owner			
QA Lead			
Project Sponsor			

This BA guide is aligned with H2O Online as implemented: mobile (`mobile/`), admin (`admin/`), backend API (`backend/`), and project documentation in `docs/` .